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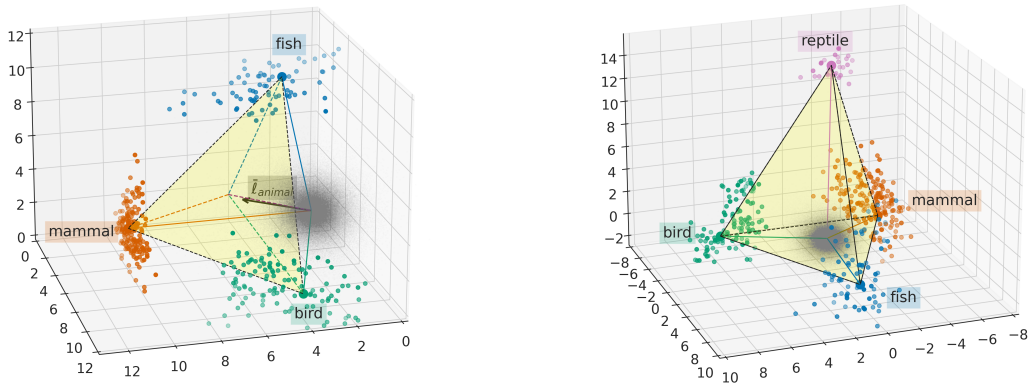

Figure 6: Categorical concepts are represented as polytopes. The plots show the projection of the unembedding vectors on the 3D subspaces: $\text{span}\{\bar{\ell}_{\texttt{mammal}}, \bar{\ell}_{\texttt{bird}}, \bar{\ell}_{\texttt{fish}}\}$ (left) and $\text{span}\{\bar{\ell}_{\texttt{bird}} - \bar{\ell}_{\texttt{mammal}}, \bar{\ell}_{\texttt{fish}} - \bar{\ell}_{\texttt{mammal}}, \bar{\ell}_{\texttt{reptile}} - \bar{\ell}_{\texttt{mammal}}\}$ (right). The gray points indicate all 256K tokens in the vocabulary, and the colored points are the tokens in $\mathcal{Y}(\texttt{w})$. The left plot further shows the orthogonality between the triangle and the projection of $\bar{\ell}_{\texttt{animal}}$ (black arrow).

A Visualization of `animal`

As a concrete example for visualizations, we examine the theoretical predictions for the familiar concept `animal`. However, the categories in WordNet are not the most intuitive; for example, the subcategories under textttanimal.n.01 include terms like "chordate", "aquatic vertebrate", "invertebrate" instead of more familiar categories such as "fish", "amphibian", or "insect". This makes it challenging to gather tokens corresponding to clear, high-level concepts $\{\texttt{mammal}, \texttt{bird}, \texttt{fish}, \texttt{reptile}, \texttt{amphibian}, \texttt{insect}\}$. Therefore, we generated two sets of tokens $\mathcal{Y}(\texttt{animal})$ and $\mathcal{Y}(\texttt{plant})$ using ChatGPT-4 (OpenAI, 2023), and manually inspected them. $\mathcal{Y}(\texttt{animal})$ is further divided into six sets of tokens for each subcategory $\{\texttt{mammal}, \texttt{bird}, \texttt{fish}, \texttt{reptile}, \texttt{amphibian}, \texttt{insect}\}$.

Figure 2 illustrates the geometric relationships between various representation vectors. The main takeaway is that the semantic hierarchy is encoded as orthogonality in the manner predicted by Theorem 8. The figure also illustrates Theorem 4, showing that the projection of the unembedding vectors for $y \in \mathcal{Y}(w)$ is approximately constant, while the projection of $y \notin \mathcal{Y}(w)$ is zero.

Figure 6 illustrates that the representation of a categorical concept is a polytope. and the projection of unembedding vectors onto the subspace for the polytope are concentrated on each vertex as predicted in Corollary 10. The left plot also shows that, as predicted, the polytope for $\{\texttt{fish}, \texttt{mammal}, \texttt{bird}\}$ is orthogonal to the vector representation of `animal`.

B Proofs

B.1 Proof of Theorem 4

Theorem 4 (Magnitudes of Linear Representations)**.** *Suppose there exists a linear representation (normalized direction) $\bar{\ell}_W$ of a binary feature W for an attribute w. Then, there is a constant $b_w > 0$ and a choice of unembedding space origin $\bar{\gamma}_0^w$ in eq.* (2.2) *such that*

$$\begin{cases} \bar{\ell}_W^\top g(y) = b_w & \text{if } y \in \mathcal{Y}(w) \\ \bar{\ell}_W^\top g(y) = 0 & \text{if } y \notin \mathcal{Y}(w). \end{cases} \tag{4.1}$$

Further, if there exist d attributes $\{w_0, \ldots, w_{d-1}\}$ such that the linear representations of the binary features for these attributes are linearly independent, we can choose a canonical origin $\bar{\gamma}_0$ in eq. (2.2)*.*

Proof. For any $y_1, y_0 \in \mathcal{Y}(w)$ or $y_1, y_0 \notin \mathcal{Y}(w)$, let Z be a binary concept where $\mathcal{Y}(Z=0) = \{y_0\}$ and $\mathcal{Y}(Z=1) = \{y_1\}$. Since Z is subordinate to W, eq. (3.2) implies that

$$\operatorname{logit} \mathbb{P}(Y = y_1 \mid Y \in \{y_0, y_1\}, \ell + \bar{\ell}_W) = \operatorname{logit} \mathbb{P}(Y = y_1 \mid Y \in \{y_0, y_1\}, \ell) \tag{B.1}$$

$$\iff \bar{\ell}_W^\top (g(y_1) - g(y_0)) = \bar{\ell}_W^\top A(\gamma(y_1) - \gamma(y_0)) = 0 \tag{B.2}$$

where A is the invertible matrix in eq. (2.2). This means that $\bar{\ell}_W^\top A\gamma(y)$ is the same for all $y \in \mathcal{Y}(w)$, and it is also the same for all $y \notin \mathcal{Y}(w)$.

Furthermore, for any $y_1 \in \mathcal{Y}(w)$ and $y_0 \notin \mathcal{Y}(w)$, eq. (3.1) implies that

$$\operatorname{logit} \mathbb{P}(Y = y_1 \mid Y \in \{y_0, y_1\}, \ell + \bar{\ell}_W) > \operatorname{logit} \mathbb{P}(Y = y_1 \mid Y \in \{y_0, y_1\}, \ell) \tag{B.3}$$

$$\iff \bar{\ell}_W^\top (g(y_1) - g(y_0)) = \bar{\ell}_W^\top A(\gamma(y_1) - \gamma(y_0)) > 0. \tag{B.4}$$

Thus, by setting $b_w^0 = \bar{\ell}_W^\top A\gamma(y)$ for any $y \notin \mathcal{Y}(w)$, and $b_w = \bar{\ell}_W^\top A\gamma(y_1) - \bar{\ell}_W^\top A\gamma(y_0) > 0$ for any $y_1 \in \mathcal{Y}(w)$ and $y_0 \notin \mathcal{Y}(w)$, we get

$$\begin{cases} \bar{\ell}_W^\top A\gamma(y) = b_w^0 + b_w & \text{if } y \in \mathcal{Y}(w) \\ \bar{\ell}_W^\top A\gamma(y) = b_w^0 & \text{if } y \notin \mathcal{Y}(w). \end{cases} \tag{B.5}$$

Then, we can choose an origin as

$$\bar{\gamma}_0^w = b_w^0 A^{-1} \bar{\ell}_W \tag{B.6}$$

satisfying eq. (4.1).

On the other hand, if there exist d attributes $\{w_0, \ldots, w_{d-1}\}$ such that the linear representations $\bar{\ell}_{W_0}, \ldots, \bar{\ell}_{W_{d-1}}$ of the binary features $W_0, \ldots, W_{d-1}$ for these attributes are linearly independent, then the linear system

$$\bar{\ell}_{W_i}^\top A\bar{\gamma}_0 = b_{w_i}^0 \quad \text{for} \quad i = 0, \ldots, d-1 \tag{B.7}$$

has a unique solution $\bar{\gamma}_0$. We can choose this vector as the canonical origin $\bar{\gamma}_0$ in eq. (2.2), ensuring that eq. (4.1) is satisfied. □

B.2 Proof of Theorem 8

Theorem 8 (Hierarchical Orthogonality)**.** *Suppose there exist the vector representations for all the following binary features. Then, we have that*

(a) $\bar{\ell}_w \perp \bar{\ell}_z - \bar{\ell}_w$ *for* $z \prec w$*;*

(b) $\bar{\ell}_w \perp \bar{\ell}_{z_1} - \bar{\ell}_{z_0}$ *for* $Z \in_R \{z_0, z_1\}$ *subordinate to* $W \in_R \{\texttt{not_w}, \texttt{is_w}\}$*;*

(c) $\bar{\ell}_{w_1} - \bar{\ell}_{w_0} \perp \bar{\ell}_{z_1} - \bar{\ell}_{z_0}$ *for* $Z \in_R \{z_0, z_1\}$ *subordinate to* $W \in_R \{w_0, w_1\}$*; and*

(d) $\bar{\ell}_{w_1} - \bar{\ell}_{w_0} \perp \bar{\ell}_{w_2} - \bar{\ell}_{w_1}$ *for* $w_2 \prec w_1 \prec w_0$*.*

Proof. (a) For $\bar{\ell}_w$ and $\bar{\ell}_z$ where $z \prec w$, by Theorem 4, we have

$$\begin{cases} (\bar{\ell}_z - \bar{\ell}_w)^\top g(y) = b_z - b_w & \text{if } y \in \mathcal{Y}(z) \\ (\bar{\ell}_z - \bar{\ell}_w)^\top g(y) = 0 - b_w = -b_w & \text{if } y \in \mathcal{Y}(w) \setminus \mathcal{Y}(z) \\ (\bar{\ell}_z - \bar{\ell}_w)^\top g(y) = 0 - 0 = 0 & \text{if } y \notin \mathcal{Y}(w). \end{cases} \tag{B.8}$$

When $w \setminus z$ denotes an attribute defined by $\mathcal{Y}(w) \setminus \mathcal{Y}(z)$, $\bar{\ell}_z - \bar{\ell}_w$ can change the target concept $w \setminus z \Rightarrow z$ without changing any other concept subordinate or causally separable to the target concept. Thus, $\bar{\ell}_z - \bar{\ell}_w$ is the linear representation $\bar{\ell}_{w \setminus z \Rightarrow z}$. This concept means $\texttt{not_z} \Rightarrow \texttt{is_z}$ conditioned on w, and hence it is subordinate to w.

Therefore, $\bar{\ell}_w$ is orthogonal to the linear representation $\bar{\ell}_{w \setminus z \Rightarrow z} = \bar{\ell}_z - \bar{\ell}_w$ by the property of the causal inner product. If they are not orthogonal, adding $\bar{\ell}_w$ can change the other concept $w \setminus z \Rightarrow z$, and it is a contradiction.

(b) By the above result (a), $\bar{\ell}_w^\top(\bar{\ell}_{z_1} - \bar{\ell}_w) = \bar{\ell}_w^\top(\bar{\ell}_{z_0} - \bar{\ell}_w) = 0$. Therefore, $\bar{\ell}_w^\top(\bar{\ell}_{z_1} - \bar{\ell}_{z_0}) = 0$.